

SUMMARY OF PRODUCT CHARACTERISTICS

1 NAME OF THE MEDICINAL PRODUCT

Rabipur

Powder and solvent for solution for injection in pre-filled syringe.

Rabies vaccine (inactivated).

2 QUALITATIVE AND QUANTITATIVE COMPOSITION

After reconstitution, 1 vial (1.0 ml) contains:

Rabies virus^a (Inactivated, strain Flury LEP)..... 27.9 IU^b

^a produced on purified chick embryo cells (PCEC)

^b quantity measured according to an ELISA

This vaccine contains residues of chicken proteins (e.g., ovalbumin), human serum albumin, and may contain traces of neomycin, chlortetracycline and amphotericin B. See section 4.4.

For the full list of excipients, see section 6.1.

3 PHARMACEUTICAL FORM

Powder and solvent for solution for injection in pre-filled syringe.

The powder is white.

The solvent is clear and colourless.

4 CLINICAL PARTICULARS

4.1 Therapeutic indications

Rabipur is indicated for active immunisation against rabies in individuals of all ages. (see sections 4.2 and 5.1)

Rabipur should be used in accordance with official recommendations.

4.2 Posology and method of administration

Posology

The recommended dose for both primary immunisation and boosters is 1.0 ml.

Pre-exposure prophylaxis

Primary immunisation

The primary pre-exposure immunisation course consists of three doses administered according to the conventional or rapid regimen, as shown in Table 1.

Table 1 Primary immunisation regimens

	<i>Conventional regimen</i>	<i>Rapid Regimen*</i>
1 st dose	Day 0	Day 0
2 nd dose	Day 7	Day 3
3 rd dose	Day 21 (or 28)	Day 7

*The rapid regimen should only be considered for adults aged 18-65 years not able to complete the conventional pre-exposure prophylaxis regimen within 21 or 28 days before protection is required.

Alternatively, in immunocompetent individuals, a one-week regimen with 2 doses can be used: at D0 and at D7.

Booster doses

Booster doses are generally recommended every 2-5 years. Timing for booster after vaccination with rapid regimen has not yet been established (see section 5.1). Serological testing for the presence of antibody ≥ 0.5 IU/ml to assess the need for booster doses should be conducted in accordance with official recommendations.

Rabipur may be used to boost individuals previously immunised with any human diploid cell rabies vaccine.

Post-exposure prophylaxis

Post-exposure prophylaxis should commence as soon as possible after exposure.

Table 2 summarises recommendations for post-exposure prophylaxis, including immunization, according to the type of exposure.

Table 2: Recommended post-exposure prophylaxis according to type of exposure

Category of exposure	Type of exposure to a domestic or wild ^{a)} animal suspected or confirmed to be rabid, or animal unavailable for testing	Recommended post-exposure prophylaxis
I	Touching or feeding animals Licks on intact skin Contact of intact skin with secretions or excretions of a rabid animal or human case	None, if reliable case history is available.

II	Nibbling of uncovered skin Minor scratches or abrasions without bleeding	Administer vaccine immediately ^{b)} Stop treatment if animal remains healthy throughout an observation period of 10 days ^{c)} or is proven to be negative for rabies by a reliable laboratory using appropriate diagnostic techniques.
III	Single or multiple transdermal bites ^{d)} or scratches, licks on broken skin. Contamination of mucous membrane with saliva (i.e. licks). Exposure to bats ^{e)} .	Administer rabies vaccine immediately, and rabies immunoglobulin, preferably as soon as possible after initiation of post-exposure prophylaxis. Rabies immunoglobulin can be injected up to 7 days after first vaccine dose administration. Stop treatment if animal remains healthy throughout an observation period of 10 days or is proven to be negative for rabies by reliable laboratory using appropriate diagnostic techniques

^{a)} Exposure to rodents, rabbits or hares does not routinely require rabies post-exposure prophylaxis.

^{b)} If an apparently healthy dog or cat in, or from a low-risk area is placed under observation, treatment may be delayed.

^{c)} This observation period applies only to dogs and cats. Except for threatened or endangered species, other domestic and wild animals suspected of being rabid should be euthanized and their tissues examined for the presence of rabies antigen by appropriate laboratory techniques.

^{d)} Bites especially on the head, neck, face, hands and genitals are category III exposures because of the rich innervation of these areas.

^{e)} Post-exposure prophylaxis should be considered when contact between a human and a bat has occurred, unless the exposed person can rule out a bite or scratch or exposure of a mucous membrane.

In-post-exposure prophylaxis of previously unvaccinated individuals, the vaccine should be administered according to Table 3.

Table 3: Post-exposure immunisation regimens for previously unvaccinated individuals

	Essen regimen (5 doses)	Zagreb regimen (4 doses)	Reduced Essen regimen (4 doses) ²
1 st dose	Day 0	Day 0, 2 doses ¹	Day 0
2 nd dose	Day 3		Day 3
3 rd dose	Day 7	Day 7	Day 7
4 th dose	Day 14	Day 21	Day 14
5 th dose	Day 28		

¹ one injection in each of the two deltoids or thigh sites

² this shortened Essen regimen may be used as an alternative for healthy, immunocompetent individuals provided they receive wound care plus rabies immunoglobulin in category III as well as in category II exposures

In previously vaccinated individuals, post-exposure prophylaxis consists of two doses administered on days 0 and 3. Rabies immunoglobulin is not indicated in such cases.

Special populations

Immunocompromised individuals

Pre-exposure prophylaxis (PrEP)

The conventional 3-dose regimen should be followed. The rapid regimen and the one-week schedule with 2 doses on days 0 and 7 may be administered, if accompanied by serological testing at 2-4 weeks after the first rabies vaccine administration to assess whether an additional vaccine administration is needed. Consultation with an infectious disease specialist or an immunologist is advised.

Post-exposure prophylaxis (PEP)

In immunocompromised individuals with category II and III exposures, 5 doses should be given in combination with comprehensive wound management and local infiltration of rabies immunoglobulin as shown in Table 4.

Table 4: Post-exposure immunisation regimens for immunocompromised individuals

	<i>Essen regimen</i>	<i>Alternative to Essen</i>
1 st dose	Day 0	Day 0, 2 doses ¹
2 nd dose	Day 3	Day 3
3 rd dose	Day 7	Day 7
4 th dose	Day 14	Day 14
5 th dose	Day 28	Day 28

¹ Two doses of vaccine may be given on day 0, that is, a single dose of 1.0 ml vaccine should be injected into the right deltoid and another single dose into the left deltoid muscle. In small children, one dose should be given into the anterolateral region of each thigh. This would result in a total of 6 doses.

When feasible, the rabies virus neutralising antibody response should be measured 2 to 4 weeks (preferably on day 14) following the start of vaccination to assess the possible need for an additional dose of the vaccine. Immunosuppressive agents should not be administered during postexposure therapy unless essential for the treatment of other conditions (see section 4.5).

Paediatric population

Paediatric individuals should receive the same dose as adults (1.0 ml).

Method of administration

Rabipur is for intramuscular administration only. For adults and children ≥ 2 years of age, the vaccine should be administered into the deltoid muscle. For children < 2 years, the anterolateral area of the thigh is recommended.

For instructions on reconstitution of the vaccine before administration, see section 6.6.

4.3 Contraindications

Pre-exposure prophylaxis (PrEP)

History of a severe hypersensitivity reaction to the active substance, to any of the excipients listed in section 6.1 or to any of the residues in section 2.

Vaccination should be postponed in individuals with a severe febrile illness (see section 4.4).

Post-exposure prophylaxis (PEP)

In view of the almost invariably fatal outcome of rabies, there is no contraindication to post-exposure prophylaxis.

4.4 Special warnings and precautions for use

A protective immune response may not be elicited in all vaccinees.

In case of acute diseases requiring treatment, patients should not be vaccinated until at least 2 weeks after recovery. The presence of a minor infection should not result in the deferral of vaccination.

Hypersensitivity reactions (PEP only)

Anaphylactic reactions including anaphylactic shock have occurred following Rabipur vaccination. As with all injectable vaccines, appropriate medical treatment and supervision should always be readily available in case of a rare anaphylactic event following the administration of the vaccine. Rabipur contains the excipient polygeline, residues of chicken proteins (e.g., ovalbumin), human serum albumin, and may contain traces of antibiotics (see section 2). In instances in which individuals have developed clinical symptoms of anaphylaxis such as generalised urticaria, upper airway (lip, tongue, throat, laryngeal or epiglottal) oedema, laryngeal spasm or bronchospasm, hypotension or shock, following exposure to any of these substances, the vaccination should only be administered by personnel with the capability and facilities to manage anaphylaxis post-vaccination.

Central nervous system effects

Encephalitis and Guillain-Barré syndrome have been temporally associated with the use of Rabipur (see section 4.8). A patient's risk of developing rabies must be carefully considered, before deciding to discontinue immunisation.

Route of administration

Rabies vaccine must not be given by intra-gluteal injection or subcutaneously, as the induction of an adequate immune response may be less reliable.

Unintentional intravascular injection may result in systemic reactions, including shock. Do not inject intravascularly.

Anxiety-related reactions

Anxiety-related reactions, including vasovagal reactions (syncope), hyperventilation or stress-related reactions, may occur in association with vaccination as a psychogenic response to the needle injection (see section 4.8). It is important that procedures are in place to avoid injury from fainting.

4.5 Interaction with other medicinal products and other forms of interaction

Immunosuppressive agents can interfere with the development of an adequate response to the rabies vaccine. Therefore, it is recommended that serological responses should be monitored in such subjects, and additional doses administered as necessary (see section 4.2).

The vaccine must not be mixed in the same syringe with other medicinal products. If rabies immunoglobulin is indicated in addition to Rabipur vaccine, it must be administered at an anatomical site distant to the vaccination.

Available clinical data support concomitant administration of Rabipur with inactivated Japanese encephalitis (JE) vaccine and conjugated MenACWY meningococcal vaccine in adult subjects; limited data are available in the paediatric population.

Almost all adult subjects achieved an adequate immune response (Rabies Viral Neutralizing Antibodies (RVNAs) ≥ 0.5 IU/ml) within 7 days after the end of a primary series of three injections of Rabipur when given concomitantly with inactivated JE vaccine according to either a rapid or the conventional PrEP schedule by the intramuscular route. From day 57 after vaccination a faster decline in immune response to rabies was observed in individuals vaccinated concomitantly with JE vaccine according to the rapid PrEP schedule compared with the concomitant conventional PrEP schedule and the rabies only conventional PrEP schedule. At day 366, percentages of subjects with RVNA concentration ≥ 0.5 IU/mL were 68%, 76%, and 80% for vaccine groups rabies/JE accelerated, rabies/JE conventional, and rabies conventional, respectively.

All adult subjects achieved an adequate immune response (RVNAs ≥ 0.5 IU/ml) within 28 days after the end of a primary series of three injections of Rabipur when given concomitantly with conjugated MenACWY vaccine according to the recommended conventional schedule by the intramuscular route.

Concomitant vaccines should always be administered at separate injection sites and preferably contralateral limbs.

4.6 Fertility, pregnancy and lactation

Pregnancy

No cases of harm attributable to use of Rabipur during pregnancy have been observed.

Rabipur may be administered to pregnant women when post-exposure prophylaxis is required.

The vaccine may also be used for pre-exposure prophylaxis during pregnancy if it is considered that the potential benefit outweighs any possible risk to the fetus.

Breastfeeding

While it is not known whether Rabipur enters breast milk, no risk to the breast-feeding infant has been identified. Rabipur may be administered to breastfeeding women when post-exposure prophylaxis is required.

The vaccine may also be used for pre-exposure prophylaxis in breastfeeding women if it is considered that the potential benefit outweighs any possible risk to the infant.

Fertility

Non clinical reproductive and developmental toxicity studies have not been performed.

4.7 Effects on ability to drive and use machines

Some of the adverse effects described in section 4.8, may affect the ability to drive and use machines.

4.8 Undesirable effects

Summary of the safety profile

Anaphylactic reactions including anaphylactic shock that are very rare but clinically severe, and potentially lethal, systemic allergic reactions, can occur following Rabipur vaccination. Mild allergic reactions to Rabipur (i.e. hypersensitivity), including rashes (very common) and urticaria (common) may occur after vaccination. These reactions are usually mild in nature and typically resolve within a few days.

Very rare cases with symptoms of Encephalitis and Guillain-Barré Syndrome have been reported following Rabipur vaccination.

In clinical trials, the most commonly reported solicited adverse reactions were injection site pain (30-85%) or injection site induration (15-35%). Most injection site reactions were not severe and resolved within 24 to 48 hours.

Tabulated list of adverse reactions

Adverse reactions considered as being at least possibly related to vaccination have been categorised by frequency.

Frequencies are defined as follows:

Very common:	($\geq 1/10$)
Common:	($\geq 1/100$ to $< 1/10$)
Uncommon:	($\geq 1/1,000$ to $< 1/100$)
Rare:	($\geq 1/10,000$ to $< 1/1,000$)
Very rare:	($< 1/10,000$)

Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

In addition to reports in clinical trials, worldwide voluntary reports of adverse reactions received for Rabipur since market introduction are included in the list. These reactions are reported voluntarily from a population of uncertain size and have been chosen for inclusion due to their seriousness, frequency of reporting, causal relationship to Rabipur, or a combination of these factors.

Table 5: Adverse reactions reported in clinical trials and in postmarketing surveillance

System Organ Class	Frequency	Adverse events
Blood and lymphatic system disorders	Common	Lymphadenopathy
Immune system disorders	Rare	Hypersensitivity
	Very rare	Anaphylaxis including anaphylactic shock*
Metabolism and nutrition disorder	Common	Decreased appetite
Nervous system disorders	Very common	Headache, Dizziness
	Rare	Paraesthesia
	Very rare	Encephalitis*, Guillain-Barré syndrome*, Presyncope*, Syncope*, Vertigo*
Gastrointestinal disorders	Common	Nausea, Vomiting, Diarrhoea, Abdominal pain/ discomfort
Skin and subcutaneous tissue disorders	Very common	Rash
	Common	Urticaria
	Rare	Hyperhidrosis (sweating)
	Very rare	Angioedema*
Musculoskeletal and connective tissue disorders	Common	Myalgia, Arthralgia
General disorder and administration site conditions	Very common	Injection site reactions, Malaise, Fatigue, Asthenia, Fever
	Rare	Chills

*Additional adverse reactions from spontaneous reporting

Paediatric population

Frequency, type and severity of adverse reactions in children are expected to be the same as in adults.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the Yellow Card Scheme at: www.mhra.gov.uk/yellowcard or search for MHRA Yellow Card in the Google Play or Apple App Store.

4.9 Overdose

No symptoms of overdose are known.

5 PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: ATC code: J07B G01

Mechanism of action

Rabipur induces stimulation of lymphocytes and antibody-secreting plasmocytes resulting in production of RVNAs.

Clinical efficacy and safety

Pre-exposure prophylaxis

In clinical trials with previously unimmunised subjects almost all subjects achieved an adequate immune response (RVNAs ≥ 0.5 IU/ml) 3 to 4 weeks after the end of a primary series of three injections of Rabipur when given according to the recommended schedule by the intramuscular route.

Persistence of adequate immune response (RVNAs ≥ 0.5 IU/ml) for up to 2 years after primary immunisation with Rabipur without booster doses has been found in clinical studies. As antibody concentrations slowly decrease, booster doses may be required to maintain antibody levels above 0.5 IU/ml.

The timing of booster doses after primary vaccination with rapid regimen or after concomitant vaccination has not yet been established; Due to a faster decline in immune response compared with the conventional schedule a shorter interval between primary vaccination and booster administration may be needed compared with the conventional vaccine schedule (see section 4.2).

In a clinical trial, a booster dose of Rabipur administered 1 year after primary immunisation elicited a 10-fold or higher increase in Geometric Mean Concentrations (GMCs) by day 30. It has also been demonstrated that individuals who had previously been immunised with Human Diploid Cell Vaccine (HDCV) developed a rapid anamnestic response when boosted with Rabipur.

Post-exposure prophylaxis

In clinical studies Rabipur elicited adequate neutralising antibodies (≥ 0.5 IU/ml) in almost all subjects by day 14 or 30, when administered according to the 5- dose (day 0, 3, 7, 14, 28; 1.0 ml each, intramuscular) Essen regimen or 4-dose (day 0 [2 doses], 7, 21; 1.0 ml each, intramuscular) Zagreb regimen.

Concomitant administration of Human Rabies Immunoglobulin with the first dose of rabies vaccine caused a slight decrease in GMCs (Essen regimen). However, this was not considered to be clinically relevant.

5.2 Pharmacokinetic properties

Not applicable

5.3 Preclinical safety data

Preclinical data including single-dose, repeated dose and local tolerance studies revealed no unexpected findings and no target organ toxicity. No genotoxicity, carcinogenicity and reproductive toxicity studies have been performed.

6 PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Powder:

Trometamol

Sodium chloride

Disodium edetate

Potassium-L-glutamate

Polygeline

Sucrose

Solvent:

Water for injection

6.2 Incompatibilities

In the absence of compatibility studies, Rabipur must not be mixed in the same syringe with other medicinal products.

6.3 Shelf life

48 months

After reconstitution the vaccine is to be used immediately.

6.4 Special precautions for storage

Store in a refrigerator (2°C - 8°C). Do not freeze.

Keep the vial and the syringe in the outer carton in order to protect from light.

For storage conditions after reconstitution of the medicinal product, see section 6.3.

The vaccine should not be used after the expiration date given on package and container.

6.5 Nature and contents of container

Package with:

1 vial (type I glass) of freeze-dried vaccine with stopper (chlorobutyl or bromobutyl)

1 disposable pre-filled syringe (type I glass) of sterile diluent for reconstitution (1 mL)

- either with a white tamper evident seal cap (polypropylene), a tip-cap (bromobutyl) and a plunger-stopper (bromobutyl)
- or with a transparent screw cap (polypropylene), a tip-cap (isoprene-bromobutyl) and a plunger-stopper (isoprene-bromobutyl)

2 identical needles (25 gauge, 25 mm) - one for reconstitution and one for administration.

Packs of:

- 1 vial and 1 pre-filled syringe with white tamper evident seal cap with needles,
- 1 vial and 1 pre-filled syringe with transparent screw cap with needles.

Not all packs may be marketed.

6.6 Special precautions for disposal

There are two types of pre-filled syringes which are differentiated by their type of cap.

Both types of syringes are equipped with a backstop which facilitates injection handling and administration. In addition, the backstop reduces the opening diameter of the syringe body and simultaneously enlarges the finger flange with ergonomically

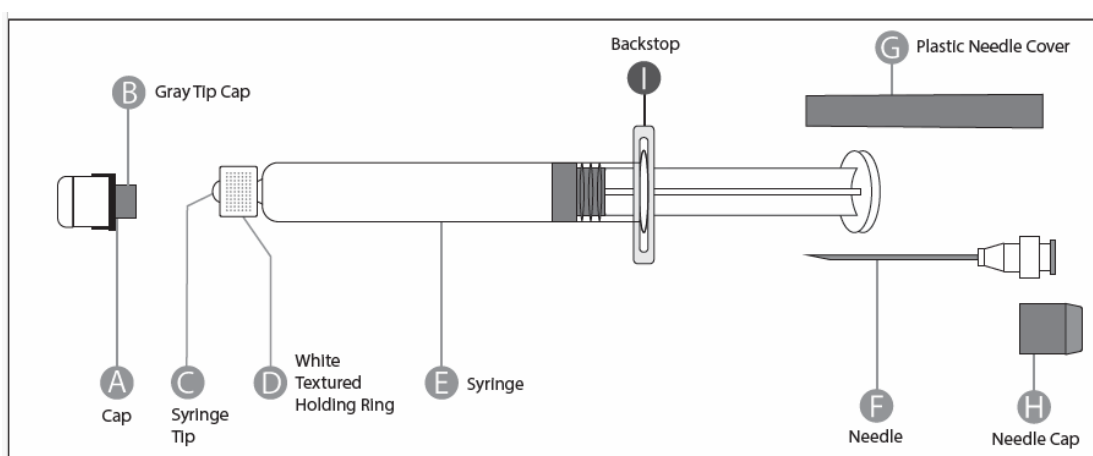
shaped wings. This prevents the plunger stopper from being inadvertently pulled out of the syringe.

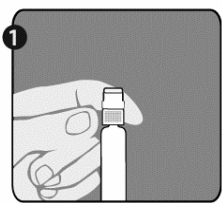
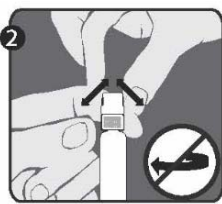
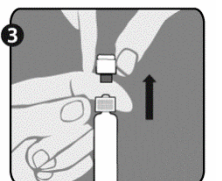
Before preparing Rabipur for administration, identify the pre-filled syringe in your pack (either with white tamper evident seal cap or with transparent screw cap) and follow the instructions of the relevant pre-filled syringe contained in the pack.

Instruction for use of Rabipur disposable pre-filled syringe:

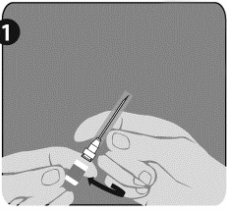
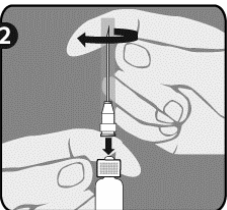


Pre-filled syringe with white tamper evident seal cap:

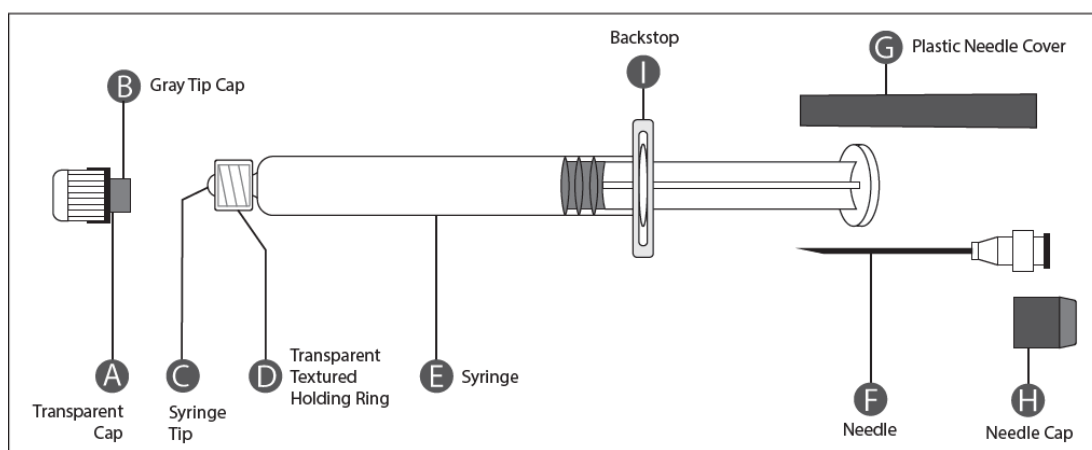


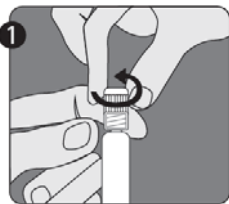
Step 1: With one hand, hold the syringe with the cap pointing upward. Be sure to hold the syringe by the white textured holding ring (D).	
Step 2: With the other hand, grasp the cap (A) and firmly rock it back and forth to break its connection to the holding ring (D). Do not twist or turn the cap.	
Step 3: Lift up to remove the cap (A) and the attached gray tip cap (B). Be careful not to touch the sterile syringe tip (C).	

Needle application (these instructions apply to both provided needles):

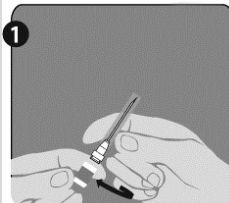
Step 1: Twist to remove the cap (H) from one of the two identical needles. This will be the needle used for reconstitution. Do not remove the plastic cover (G).	
Step 2: With one hand, firmly hold syringe by white textured holding ring (D). With your other hand, insert this needle (F) and twist clockwise until it locks into place. Once the needle is locked, remove its plastic cover (G). The syringe (E) is now ready for use.	

Pre-filled syringe with transparent screw cap:

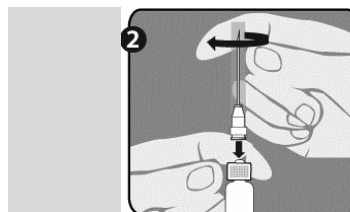


Step 1: With one hand, hold the syringe with the cap pointing upward. Be sure to hold the syringe by the transparent textured holding ring (D). Unscrew the cap (A) by twisting it counterclockwise.	
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Needle application (these instructions apply to both provided needles)

Step 1: Twist to remove the cap (H) from one of the two identical needles. This will be the needle used for reconstitution. Do not remove the plastic cover (G).	
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Step 2: With one hand, firmly hold syringe by transparent textured holding ring (D). With your other hand, insert this needle (F) and twist clockwise until slight resistance is felt. Once the needle is locked, remove its plastic cover (G). The syringe is now ready for use.



Instructions for reconstituting Rabipur with the use of pre-filled syringe:

The vaccine should be visually inspected both before and after reconstitution for any foreign particulate matter and or change in physical appearance. The vaccine must not be used if any change in the appearance of the vaccine has taken place.

The reconstituted vaccine is clear to slightly opalescent and colourless to slightly pink.

The powder for solution should be reconstituted using the solvent for solution supplied and carefully agitated prior to injection. The reconstituted vaccine should be used immediately.

During manufacturing, the vial is sealed under vacuum. Therefore to prevent problems in withdrawing the reconstituted vaccine from the vial after reconstitution of the vaccine, it is recommended to unscrew the syringe from the needle to eliminate the negative pressure. After that, the vaccine can be easily withdrawn from the vial. It is not recommended to induce excess pressure, since over-pressurization will create the problems in withdrawing the proper amount of the vaccine.

The length of the needle will not reach to the bottom of the vial, so please invert the vial and pull back the needle close to the stopper. This will allow the full amount of vaccine solution to be withdrawn from the vial.

After completing the reconstitution of the vaccine, remove the cap from the second needle (as explained in step 1 for the needles) and replace the needle used for reconstitution with the second needle, to be used for administration. Do not use the same needle for reconstitution and administration.

Any unused vaccine or waste material should be disposed of in accordance with local requirements.

7 MARKETING AUTHORISATION HOLDER

Bavarian Nordic A/S
Philip Heymans Alle 3
2900 Hellerup
Denmark

8 MARKETING AUTHORISATION NUMBER(S)

PL 40365/0004

**9 DATE OF FIRST AUTHORISATION/RENEWAL OF THE
AUTHORISATION**

06/04/2016

10 DATE OF REVISION OF THE TEXT

20/05/2026